

Lab 11

[valid 2024-2025]

REST Services

Continue the application created at [lab 8](#) or [lab 9](#) integrating the following functionalities:

- Implement REST services needed to communicate with the server side data (CRUD).

The main specifications of the application are:

Compulsory (1p)

- Create a Spring Boot project that will contain the REST services for communicating with the server data.
 - Create a REST controller containing a method for:
 - obtaining the list of countries, via a HTTP GET request.
 - Test your service using the browser and/or [Postman](#), or [RestMan](#), etc.
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Homework (2p)

- Create REST services for:
 - obtaining the list of cities, via a HTTP GET request.
 - adding a new city, via a HTTP POST request.
 - modifying the name of a city, via a HTTP PUT request.
 - deleting a city, via a HTTP DELETE request.
 - Create a simple client application that invokes the services above, using the support offered by Spring Boot.
 - Document your services using [Swagger](#) or a similar tool.
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Bonus (2p)

- Create a service that assigns a color to each country, such that two [neighboring countries](#) do not have the same color, using [a minimum number of colors](#).
- Secure this service using [JSON Web Tokens](#).

Resources

- [What is REST?](#)
- [Spring Boot Reference Documentation](#)

Objectives

- Understand the concept of application framework.
- Get familiar with Spring Boot.
- Understand REST software architectural style.
- Write programs that communicate with other programs on the network, using HTTP.